the p–V diagram.

3 **(a)** For a to be minimum, the minimum intensity detected directly below it must be the 1st minimum from the central bright fringe. i.e., $n = 0.5\lambda$

$\therefore$ Path difference = 0.5 λ

$$\sqrt{2.00^2 + a^2} - 2.00 = 0.5\lambda$$

$$a^2 + (2.00)^2 = (2.00 + 0.5\lambda)^2$$

$$a^2 + 4 = 4 + 2\lambda + 0.25\lambda^2$$

$$a = \sqrt{4\lambda + 0.25\lambda^2}$$

$$= \sqrt{(750 \times 10^{-9})(2 + 0.25(750 \times 10^{-9}))} = \sqrt{1.50 \times 10^{-6}}$$

$$= 1.22474 \times 10^{-3} = 1.22 \times 10^{-3}$$

λ has to be substituted in at the last step and not earlier as calculator has insufficient number of digits.

OR Since the interference pattern is symmetrical about perpendicular bisector between S_1 and S_2,

$$a = 0.5\Delta x + 0.5\Delta x = \Delta x$$

OR

Distance (on the screen) between the central maximum and the 1st minimum is

$$\frac{x}{2} = \frac{a}{2}$$

$$a = \frac{\lambda D}{\Delta x} = \frac{\lambda D}{a}$$

$$a^2 = \lambda D = \left(750 \times 10^{-9}\right)(2.00) = 1.50 \times 10^{-6}$$

$$a = 1.22474 \times 10^{-3} = 1.22 \times 10^{-3} \text{ m}$$

(b) **(i)**

$$v = \frac{3\Delta x}{(1.0)}$$

$$= 3\frac{\lambda D}{a}$$

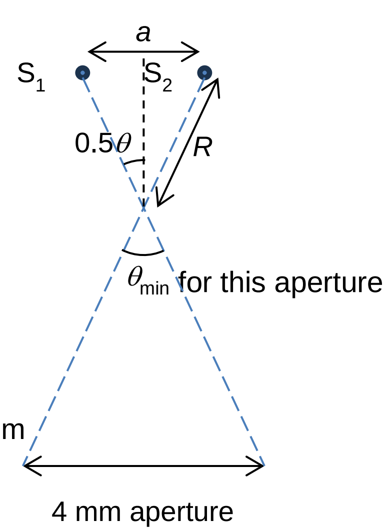
$$= \frac{3\left(750 \times 10^{-9}\right)(2.00)}{1.2247 \times 10^{-3}}$$

$$= 3.67 \times 10^{-3} \text{ m s}^{-1}$$

(ii) The frequency will decrease.

$\underline{\Delta x = \frac{\lambda D}{a}. \ D \text{ increases as the detector moves.}}$ Hence, Δx increases and the detector has to move a corresponding longer distance to detect consecutive maxima.

(c) For images to be just resolved,



$$\theta_{\text{min}} = \frac{\lambda}{b} = \frac{750 \times 10^{-9}}{4.0 \times 10^{-3}} = 1.875 \times 10^{-4} \text{ rad}$$

$$s = R\theta_{\text{min}}$$

$$R = \frac{a}{\theta_{\text{min}}} = \frac{1.22 \times 10^{-3}}{1.875 \times 10^{-4}} = 6.507 \text{ m}$$

OR

$$\tan\left(\frac{1}{2}\theta_{\text{min}}\right) = \frac{\frac{1}{2}a}{R} \quad \Rightarrow R = \frac{\frac{1}{2}\left(1.22 \times 10^{-3}\right)}{\frac{1}{2}\left(1.875 \times 10^{-4}\right)} = 6.507 \text{ m}$$

4 **(a)** The electric potential at a point in an electric field is defined as the work done per